

NeoCampus Fingerprint Terminal

Stage 5 Completion & Handover Report

Project: NeoCampus Fingerprint Device / NeoCard Terminal
Stage: Stage 5 — Final Device Completion & System Validation
Organization: Kardiverse Technologies LTD
Report date: 12 August 2026
Physical validation date: 12 August 2026
Demonstration video: Completed (recorded by delivery team)

1. Executive summary

The NeoCard fingerprint terminal on device **KDVC-RPI-001** has completed Stage 5 **core physical validation**. The end-to-end product path is proven on hardware:

Power / systemd → hardware init → device auth → TFT ready → enrollment → known fingerprint verification → NeoCard check-in → unknown rejection → offline queue → network recovery → sync

Area	Verdict
Software regression (local)	PASS
Physical E2E core path (Phase B)	PASS
Demonstration video	Completed
Open defects (non-blocking core path)	DEFECT-003 (systemd stop timeout), DEFECT-004 (cosmetic enroll display)
Optional items not re-run on Pi	P-14 reboot-with-unsynced-queue; live P-13 idempotency (software PASS)

Stage 5 is a **validation and handover** stage. No new product features were added during physical validation beyond defect fixes already recorded in Step 6.

2. System under test

Item	Value
Device ID	device_1783962666378_8d478192
Device name	KDVC-RPI-001
Assigned user	user_1784053930781_e82767e8 (Kelvin Muchemi)
Fingerprint slot	1
Enrollment ID	enroll_kelvin_slot1_20260812 (ACTIVE)
Pi application	kdvc-fingerprint (Python)
Pi software HEAD	edf0cb2 — fix(stage5): no-hardware startup & test storage isolation
Functional baseline tag	stage5-step4-baseline → 24c9d77
Backend	neocard-backend (Node/Express/SQLite)
Backend API	http://192.168.2.105:3000

systemd unit	kdvc-fingerprint.service (enabled)
Sensor	R503 @ 57600 baud (/dev/serial0)
Offline queue	~/kdvc-fingerprint/data/offline_transactions.db

Secrets (device API keys) are loaded only via `.env` / `systemd EnvironmentFile` and are **not** included in this report.

3. Software baseline (Phase A)

Suite	Result
Backend npm test	50/50 PASS
Pi software unit/integration (post Step 6)	72/72 PASS

Step 6 defect fixes (software):

Defect	Status	Summary
DEFECT-001	FIXED	--no-hardware no longer opens /dev/serial0
DEFECT-002	FIXED	Unit tests isolated from production data/store.json

4. Physical validation matrix (Phase B)

ID	Check	Result
P-01	Power ON → Raspberry Pi OS	PASS
P-02	systemd starts kdvc-fingerprint	PASS
P-03	App loads .env via EnvironmentFile	PASS
P-04	Hardware init (display, touch, RTC, buzzer, fingerprint)	PASS
P-05	Device authentication (/v1/device/me)	PASS
P-06	TFT splash → home READY / Place Finger	PASS
P-07	Enrollment (slot 1 ACTIVE in backend)	PASS
P-08	Known fingerprint → VERIFIED + check-in	PASS
P-09	NeoCard transaction visible in backend	PASS
P-10	Unknown fingerprint → UNKNOWN / no check-in	PASS
P-11	Backend/network loss → offline queue	PASS
P-12	Network recovery → SYNCED + backend SUCCESS	PASS
P-13	Replay same transaction_id (no duplicate)	PASS (software); live optional
P-14	Reboot with unsynced queue	NOT RE-RUN
P-15	journalctl usable; no secrets in logs	PASS

Physical scorecard

Field	Status
SYSTEMD_BOOT / STARTUP	PASS
ENROLLMENT	PASS
KNOWN_FINGER	PASS
UNKNOWN_FINGER	PASS
NEOCARD_CHECKIN	PASS
OFFLINE	PASS
NETWORK_RECOVERY / SYNC	PASS
SYSTEMD_CLEAN_STOP	FAIL (DEFECT-003)

Phase B core path: PASS

5. Key lab evidence (12 August 2026)

5.1 Sensor hygiene

R503 initially held **4** stale templates. Search matched wrong slots (2 / 5) while backend only had slot **1**, producing UNKNOWN. Library was cleared (`template_count: 0`), slot 1 template rewritten on sensor; existing backend ACTIVE enrollment retained.

Operational lesson: Keep sensor flash aligned with backend enrollments before demos.

5.2 Known finger

- TFT: Welcome Kelvin Muchemi / Check-In Successful
- Journal: `Verification outcome=VERIFIED ... slot=1`
- Backend: `CHECK_IN / SUCCESS` for slot 1 (example ~18:08:37 UTC)

5.3 Unknown finger

- Journal: `Verification outcome=UNKNOWN ... slot=None`
- No spurious NeoCard check-in for that attempt

5.4 Offline → sync

Transaction ID	Queued	After recovery	Backend
add9518c-298b-4de4-92bc-34b751f1e02f	OFFLINE slot 1	SYNCED	SUCCESS
29dc9cdb-4f62-43dc-8888-422f0ce0752a	OFFLINE slot 1	SYNCED	SUCCESS

Both backend SUCCESS rows recorded at approximately **18:24:21 UTC**.

6. Open defects (handover)

DEFECT-003 — systemd stop timeout → SIGKILL

- **Severity:** Medium (lifecycle / ops)
- **Impact:** Does **not** block verify / check-in / offline sync
- **Behaviour:** `systemctl stop` may hit stop-timeout and SIGKILL
- **Likely cause:** Verify loop blocked waiting for finger / UART; shutdown not interrupted promptly
- **Disposition:** Accepted as known limitation for Stage 5 handover unless a minimal lifecycle fix is scheduled

DEFECT-004 — Enrollment CLI empty response fields (cosmetic)

- Success message correct; printed User/Device/Slot may show as em dash while DB row is complete
- **Disposition:** Cosmetic; optional follow-up

7. Demonstration video

Item	Status
Video recorded	Yes (completed by delivery team, 12 August 2026)
Intended content	Ready state; unknown rejection; known verify + welcome + check-in; backend SUCCESS; offline queue; recovery / sync; return to ready
Guide used	docs/STAGE5_DEMO_VIDEO.md

The video demonstrates the finished physical product. It is a required Stage 5 MOU artefact alongside this written report.

8. Repositories & documentation index

Artefact	Location
This completion report (Markdown)	neocard-backend/docs/STAGE5_COMPLETION_REPORT.md
This completion report (PDF)	neocard-backend/docs/STAGE5_COMPLETION_REPORT.pdf
Phase B physical results	docs/STAGE5_PHASE_B_PHYSICAL_RESULTS.md
Demo video guide	docs/STAGE5_DEMO_VIDEO.md
Step 5 regression matrix	docs/STAGE5_STEP5_REGRESSION.md
Step 5 software results	docs/STAGE5_STEP5_REGRESSION_RESULTS.md
Software RC freeze	docs/STAGE5_SOFTWARE_RC.md
Stage 5 specification	docs/STAGE5_SPEC.md

Companion application repository: `kdv-c-fingerprint` .

9. Remaining optional close-out

These items do **not** invalidate the Phase B core PASS:

1. Optional P-14: reboot with one unsynced OFFLINE row
2. Optional live P-13: explicit same- `transaction_id` replay on hardware
3. Quarantine remaining historical `FAILED` offline queue rows on the Pi
4. Decide whether to schedule a minimal fix for DEFECT-003
5. Attach or archive the demo video file with client naming conventions

10. Conclusion

The NeoCampus Fingerprint Terminal (**KDVC-RPI-001**) has demonstrated the complete fingerprint-to-NeoCard workflow on physical hardware, including offline resilience and synchronization. Software regression is green. The demonstration video has been completed. Remaining items are optional or non-blocking defects documented for handover.

STAGE5_CORE_VALIDATION=PASS
PHASE_B=CORE_PASS
DEMO_VIDEO=COMPLETED
DATE=2026-08-12
DEVICE=KDVC-RPI-001 / device_1783962666378_8d478192

Prepared for Stage 5 MOU handover — Kardiverse Technologies LTD / NeoCampus Fingerprint Device.